Mapping Sugarcane Crop Age and Harvest Dynamics Using Sentinel-2 Data

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A Multi-Temporal Analysis of Sugarcane Paddocks in the Palmers Channel and Palmers Island Region, New South Wales, Australia (2016–2025)

GEOM7001 Project - October 2026
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1. Abstract

In the Northern Rivers region of New South Wales (NSW), Australia, sugarcane is grown for one or two years, compared to the one year growth cycle undertaken in the much larger growing region of Queensland. As sugarcane grows for its second year, it becomes yellower and hence has different spectral properties. Here we show how multispectral satellite data from Sentinel-2 can be used to delineate one and two year old crops in the Palmers Channel and Palmers Island region of northern NSW, with 86% accuracy. Furthermore, paddocks were classified as harvested / not harvested with 91% accuracy. The study shows that the proportion of crops harvested as one year old has remained stable throughout the last decade, despite annual fluctuations due to external factors, including environmental and economic factors. The results demonstrate the feasibility of such remote sensing workflows, however further work is needed to improve the breadth of the training data to improve the accuracy. Additional data sources, including hyperspectral, radar and lidar, could be used to further improve the workflows.

2. Introduction and Background

Remote sensing has been widely used to improve understanding of various aspects of the sugarcane agriculture sector, including mapping, yield estimation, variety and phenology. Previous work has used both multispectral and hyperspectral data, with some more recent studies using radar data as well.

Som-ard et al. (2021) provides an overview of remote sensing approaches used for sugarcane cultivation. The review analysed 100 studies published from 1980 - 2020, covering sugarcane mapping, crop growth anomaly detection, health monitoring, and yield estimation. The review found that Sentinel-2 in particular is a useful data source for sugarcane applications, due to its high spatial resolution and frequent revisit. One challenge identified by the review was the *"the diversity of environmental and growth conditions - even for a given country - under which sugarcane is grown"* (Som-ard et al., 2021). As many Australian sugarcane studies are conducted in the larger Queensland growing region, studies conducted on the smaller NSW growing region are needed.

Annual sugarcane extent across Queensland from 2004 onward was estimated in Pringle (2021), through the use of Landsat and Sentinel-2 time series, combined with image segmentation and machine learning. The end-of-autumn model achieved 93% user's accuracy and 98% producer's accuracy using 596 independent random validation points. Importantly, the study showed that sugarcane's seasonal increase in vegetation greenness

could be exploited for annual detection and that near-real-time mapping was feasible. It is this seasonal increase in vegetation greenness that this project aims to exploit, however noting that sugarcane crops in Queensland are only grown for one year, while in New South Wales they are grown for up to two years.

Only a handful of studies using remote sensing for sugarcane applications have been conducted specifically in the New South Wales growing region. Muir, Robson and Rahman (2018) studied the Harwood growing region, alongside 14 other growing regions in Queensland and New South Wales, to develop time series yield forecasting models using Sentinel-2 data. This study highlighted the need for individual models for each growing region.

Furthermore, Robson et al. (2016) compared the Bundaberg (QLD) and Condong (NSW) growing regions, with the results for Condong discriminated by year 1 and year 2 growth stages. Condong is just north of Harwood, and both regions have similar growing patterns. The paper does not specify how the growth stages were discriminated between year 1 and year 2.

2.a. Crop-cycle and variety mapping

Crop cycle mapping using remote sensing was demonstrated by Everingham et al. (2007), where EO-1 Hyperion hyperspectral imagery and statistical classifiers were used to discriminate sugarcane variety and crop cycle, including the number of ratoon crops following the original planting. The authors found that support vector machines (SVM) and random forests (RF) outperformed more traditional statistical approaches. While hyperspectral data will not be used for this project, Sentinel-2's VNIR and SWIR bands may be able to support a similar workflow.

Similarly, Singh, Patel and Danodia (2022) derived phenological metrics from dense Landsat OLI NDVI time series, smoothed using the Savitzky-Golay method and analysed with TIMESAT. A decision-tree approach was then used to distinguish planted and ratoon sugarcane. Analysis of temporal descriptors such as seasonality, amplitude and timing of vegetation development can support the determination of crop age.

2.b. Object-based Classification

Previous studies have found that object-based (rather than pixel-based) image analysis can improve classification accuracy and reduce workflow, as sugarcane fields are relatively spatially and spectrally homogeneous. Vieira et al. (2012) represented paddocks as objects and used a decision-tree classifier to exploit temporal information to map sugarcane over

large areas in São Paulo, Brazil. Furthermore, Everingham et al. (2007) showed that aggregating pixel classifications to paddock level improved the assignment of crop-cycle classes.

The work of Luciano et al. (2018) is important to consider in developing the methodology for this project. Their Remote Sensing of Environment study tested three strategies for generalising sugarcane classifiers across years: cross-year calibration, multi-year calibration, and map updating using change detection. Using eight years of Landsat imagery and random forests, the multi-year classifier achieved an average Dice coefficient of approximately 0.91 when applied to independent years, compared with approximately 0.84 for simple cross-year transfer.

2.c. Use of C-band SAR data

A number of studies have used C-band SAR data from Sentinel-1 to supplement multispectral imagery. For example, Shendryk, Davy and Thorburn (2021) used approximately 1,400 field-level observations and Sentinel-1/-2 imagery over four growing seasons (2016–2019) in the Wet Tropics. Machine-learning models predicted cane yield, commercial cane sugar, sugar yield, variety and ratoon number. The differentiation of planted versus ratoon crops was possible but comparatively weak, with balanced accuracy reaching only 45.4%. Sentinel-2 spectral indices were among the most important predictors for yield, variety and ratoon-number prediction.

Nihar et al. (2022) used Sentinel-1 and Sentinel-2 to map sugarcane and subsequently separate plant and ratoon fields. Random Forest (RF) and Support Vector Machine (SVM) classifiers were used for initial sugarcane classification, followed by phenology- and spectral-based classification of plant versus ratoon cane. This study demonstrates that Sentinel-1/-2 temporal information can support not only crop-type mapping but also differentiation of sugarcane crop-cycle classes.

The physical basis for using Sentinel-1 throughout the sugarcane growth cycle was investigated by den Besten et al. (2023). Their Remote Sensing of Environment study found that Sentinel-1 VV backscatter was related to stalk development and that all tested polarizations responded to water beneath a mature canopy. This is relevant to the study site as flooding and wet conditions regularly occur. A relative-age model should distinguish harvest/regrowth signals from environmental effects such as inundation or unusually wet soil.

The main benefit cited for using SAR data is to eliminate the issue of cloud-cover which impacts optical observations - this is referenced in Shendryk et al. (2020) and Sreedhar,

Varshney and Dhanya (2022). However, as this project does not need frequent data and the time windows for acquiring data are large, there were no issues with finding appropriate cloud free data. Hence, it was determined the use of Sentinel-1 data was out of scope for this project.

2.d. Research Gap

The literature review did not find any previous studies which classified sugarcane crops by age based on spectral information alone. A study of Palmers Channel and Palmers Island using a 2016–2025 Sentinel-2 time series could therefore contribute a locally validated workflow that identifies sugarcane, detects pre-/post-harvest transitions, assigns relative crop age between harvest events, and evaluates how consistently those age/harvest signatures transfer across years.

3. Aims and Objectives

The objective of this study is to produce a time series for 2016-2025 of one-year old vs two-year old sugar cane harvested in the Palmers Channel and Palmers Island area, using multispectral data from Sentinel-2.

This will include object-based classification of paddocks as one-year old or two-year old sugar cane crops, or non-sugar cane crop (fallow). Paddocks will then be classified as either harvested or not harvested for a particular season.

4. Data and Methodology

The below sections describe the choice of study site, dataset selection and the processing workflow developed to address the aims and objectives.

Digital Earth Australia (DEA) Sandbox was used for processing all data and producing all figures for this project, referencing notebooks from Krause et al. (2021).

4.a. Study Site

The study site includes the Palmers Channel and Palmers Island areas on the Clarence Valley floodplain in northern NSW (see Figure 1). The climate is sub-tropical, and the region sees semi-regular flooding (Bureau of Meteorology, n.d.).

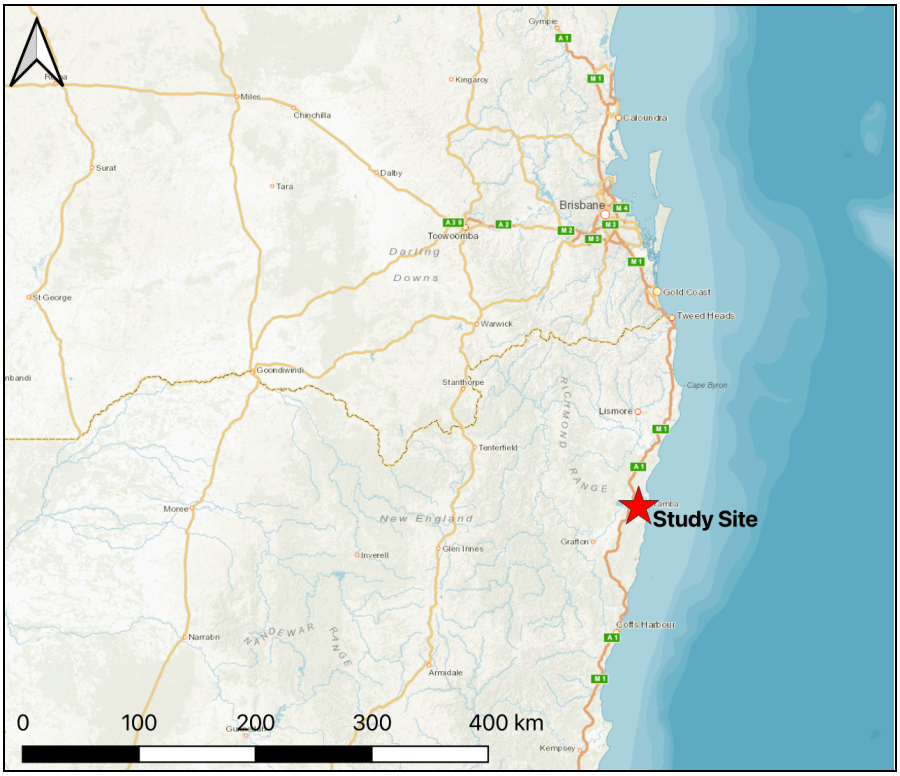


Figure 1: Location of study site in Northern NSW, plotted on the Geoscience Australia National Base Map (Geoscience Australia, 2020).

In this region, sugar cane is planted in September - November, grown for 1-2 years, then harvested between June - December. Harvests are conducted in four 'rounds', with harvesting crews visiting each farm approximately every six weeks during the harvest season (Sunshine Sugar, n.d.).

Shapefiles of the paddock outlines for the 2018 season were obtained from the Harwood Sugar Mill. As we will use Sentinel-2 data, the minimum mapping unit should be no less than 30m. Hence, the polygons in the study site were filtered to only include paddocks with dimensions greater than 30m x 30m. This leaves 491 paddocks to be included in this study. The polygons are plotted in Figure 2, overlaid on Sentinel-2 imagery from the 2018 season, pre-harvest. Figure 3 shows the farm at 130 McLeays Lane, Palmers Channel, which will be used for validation and error assessment.

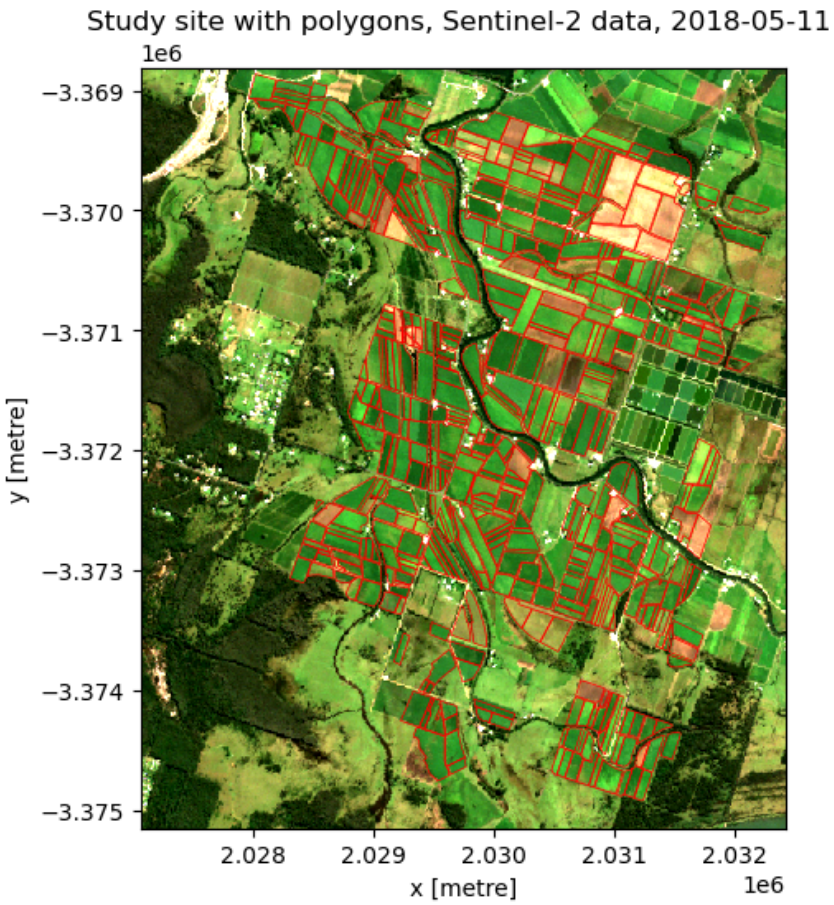


Figure 2: Sentinel-2 true colour image (Geoscience Australia, 2022) showing the study site with paddock polygons overlaid in red.

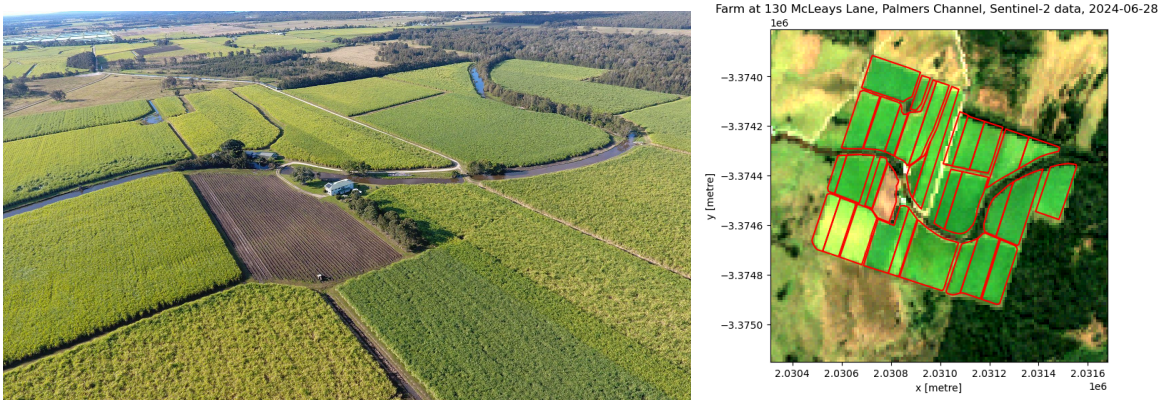


Figure 3: The farm at 130 McLeays Lane, Palmers Channel. Left - Aerial image taken July 2024, supplied by the farm owner. Right - Sentinel-2 true colour image (Geoscience Australia, 2022) with the paddock polygons overlaid.

4.b. Datasets used

Sentinel-2 Multispectral Imager (MSI) data was used for this project, specifically the DEA Surface Reflectance Nadir corrected Bidirectional reflectance distribution function Adjusted Reflectance Terrain corrected (NBART) Sentinel-2A, -2B and -2C products (Geoscience Australia, 2022). Geometric, radiometric and atmospheric corrections have been applied.

Table 1: Product specifications of the DEA Sentinel-2 NBART products.

Parameter	DEA Sentinel-2 NBART Product Specifications
Temporal Coverage and Revisit	10 day revisit from 12 Jul 2015 - 30 Jun 2017, followed by 5 day revisit (two platforms).
Spectral Bands (Spatial Resolution (m))	Coastal Aerosol (60) Blue (10) Green (10) Red (10) Red Edge 1 (20) Red Edge 2 (20) Red Edge 3 (20) NIR 1 (10) NIR 2 (20) SWIR 2 (20) SWIR 3 (20)
Radiometric Resolution	12 bit

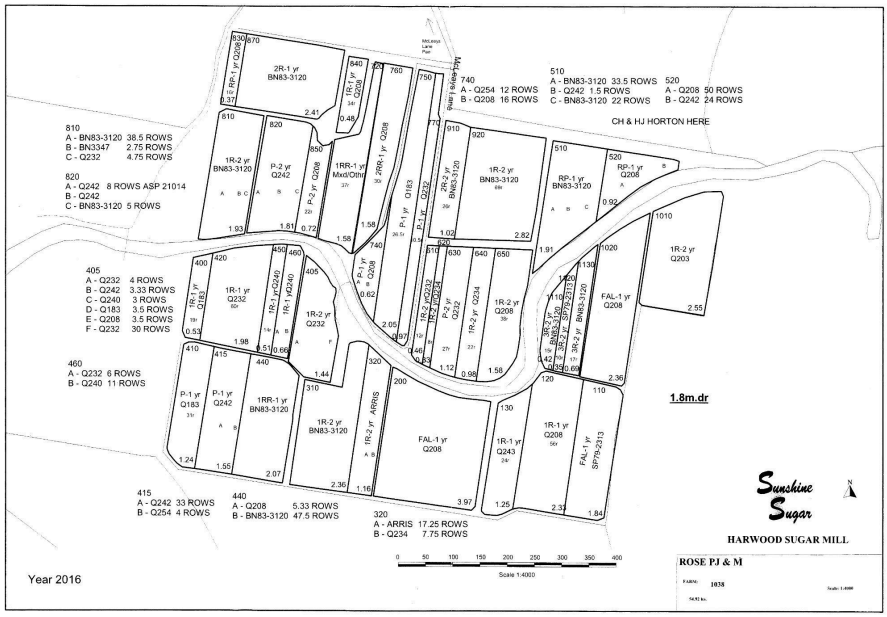


Figure 4: 2016 farm map for 130 McLeays Lane, Palmers Channel. The age is listed as '1 yr' or '2 yr'. 'FAL' indicates a fallow paddock, and is assigned age 0.

The product also contains observation attribute layers such as cloud pixels and cloud probability (s2cloudless) which can be used to filter for cloud free scenes.

For validation and error assessment, data on crop age for the years 2016 - 2023 were obtained for the farm at 130 McLeays Lane, Palmers Channel. The change between age from year to year was used to determine whether or not the paddock was harvested.

4.c. Methods Flowchart

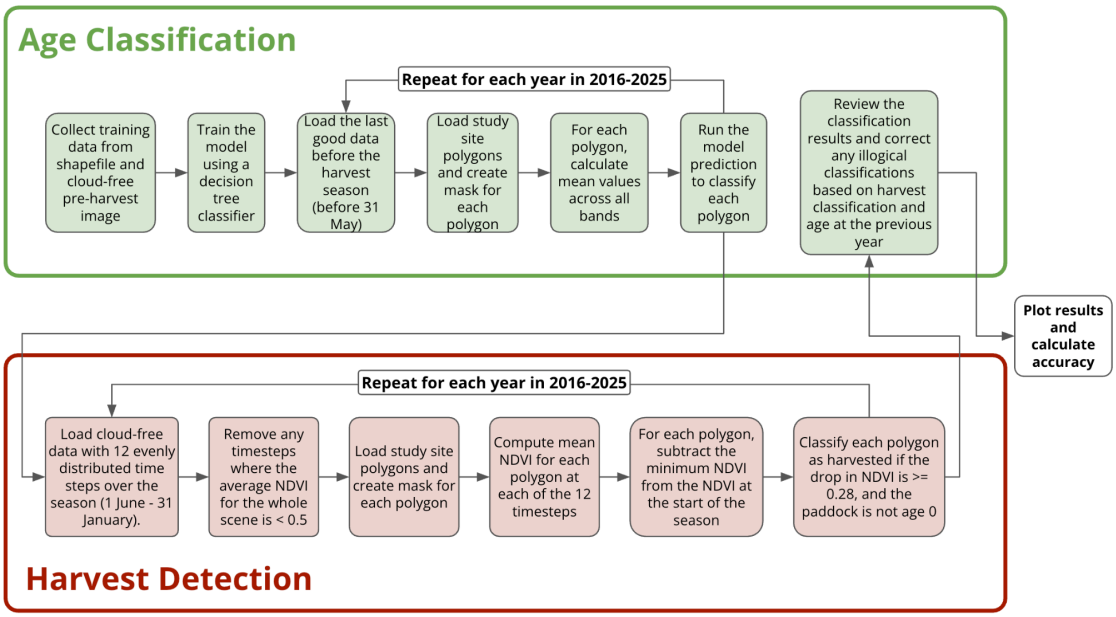


Figure 5: Data processing workflow.

The data processing workflow described in Figure 5 loaded images for training and classification using the `load_ard` function with the `s2cloudless` cloud mask, and setting a threshold for good data at 0.99. This ensures cloud free scenes.

For the age classification workflow, a decision tree classifier was used, trained on data from 2018. An assumption is made that every paddock is of a uniform age, and hence the mean value for each polygon was computed for each band. The model prediction found that the SWIR-3, Red and Red Edge 1 bands were most important to classify the ages of paddocks (see Figure 6).

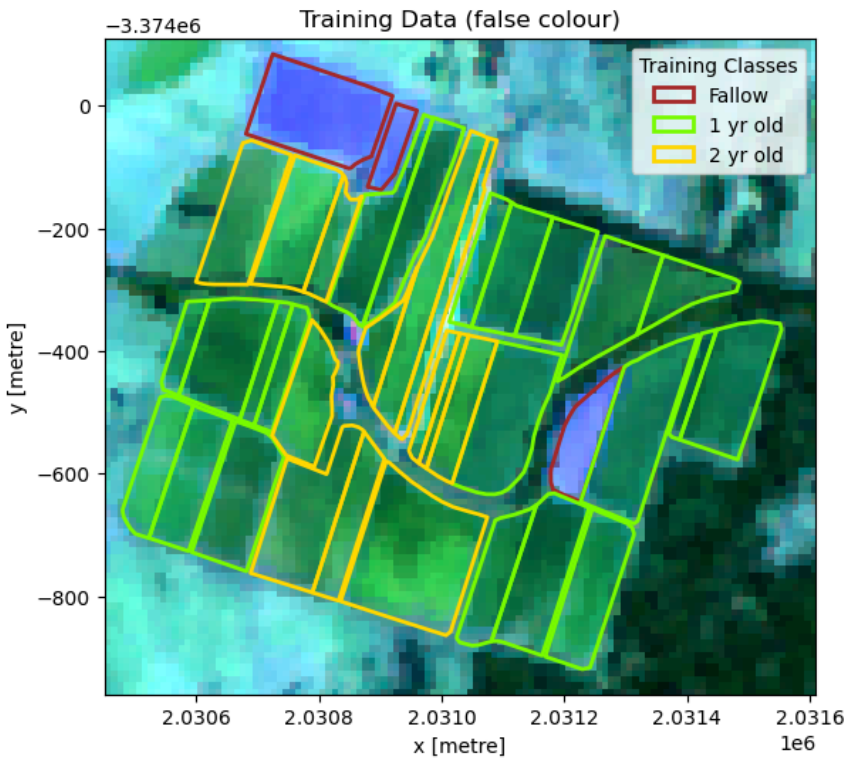


Figure 6: Sentinel-2 false colour image of training data, with Red = Red band, Green = Red Edge 1 band, Blue = SWIR-3 band.

For harvest detection, mean Normalised Difference Vegetation Index (NDVI) values were computed for each polygon, at 12 time steps throughout the season. Any timesteps where the mean NDVI across the whole image was less than 0.5 were removed, as particularly dry seasons (as seen in 2019) can result in paddocks being falsely classified as harvested. From the remaining good time steps, the minimum NDVI is calculated, and from the initial NDVI, to produce an NDVI drop value. If the NDVI drop is greater than the threshold of 0.28, the paddock is classified as harvested. The results of the age classification are also referenced, and any paddock with age 0 (fallow paddocks) are automatically classed as not harvested.

The results of the harvest classification are then passed back to the age classification workflow, and any illogical ages are adjusted. This includes that an age 2 paddock cannot have been harvested the year before (it must be instead age 1), and a paddock cannot grow more than one year between years (the age is adjusted to be the maximum possible age). This removes many false classifications of 1-year-old paddocks as 2-year-old.

Using the validation data from 130 McLeays Lane, Palmers Channel for the years 2016 - 2023 (excluding 2018 as that was used for training), confusion matrices are calculated for both the age and harvest classifications. A total of 220 and 218 data points were available

for accuracy assessment of the age classification and the harvest classification, respectively.

5. Results & Discussion

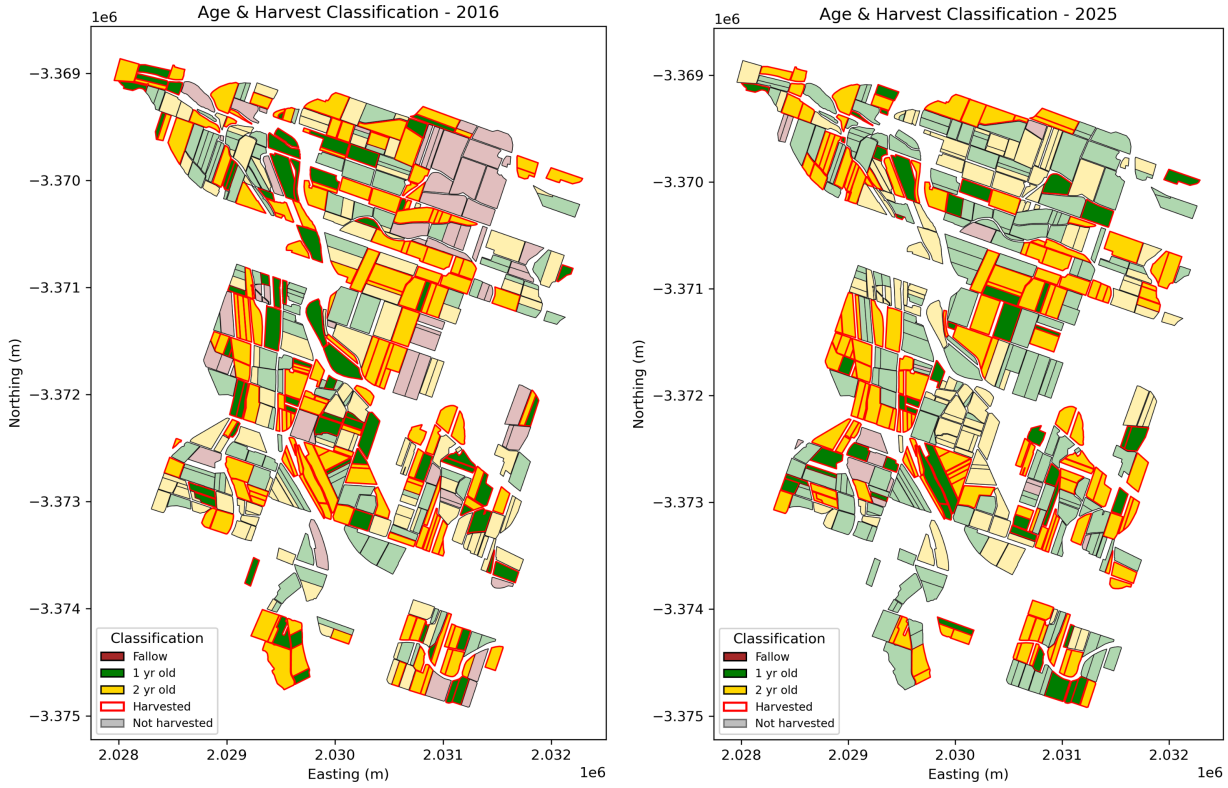


Figure 7: Age and Harvest Classification for 2016 (left) and 2025 (right) - see Appendix for full maps for all other years. Polygons outlined in red and at full opacity were classified as harvested, while those at 50% opacity were not harvested.

For each year in 2016 - 2025, maps similar to those in Figure 7 were produced (see Appendix). The overall accuracy of the age classification was 86.36%, while the harvest classification had an accuracy of 91.74%. The accuracy of the age and harvest classifications are shown in the below confusion matrices.

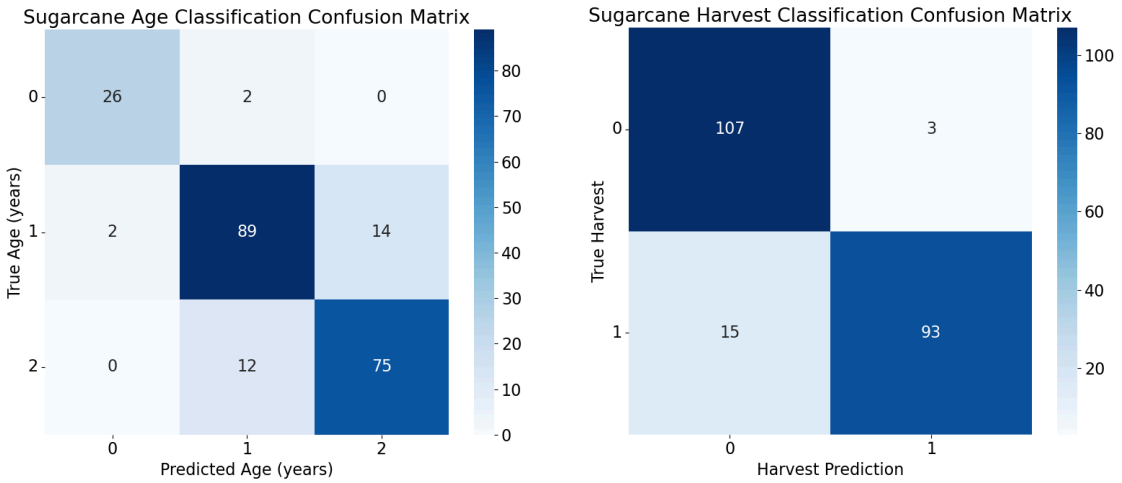


Figure 8: Age (left) and Harvest (right) Classification Confusion Matrices. For Age, 0 = Fallow, 1 = 1 year old, 2 = 2 year old. For Harvest Prediction, 0 = not harvested, 1 = harvested.

The largest sources of error in the age classification were one and two year old paddocks being mis-classified, with 12% and 16% commission error for the one and two year old classes, respectively. More timesteps throughout the year (not just the harvest season) could be analysed to better determine the growth stage of the crop and improve the accuracy of the classification.

As shown in the appendix (Figures A1 and A2), the spectral separation between the one and two year old classes is limited. The bands with the largest separation were the red, red edge 1 and NIR 1 bands, but there is still a significant amount of overlap where misclassification can occur. This is where additional information is needed to delineate the classes, such as the logic applied regarding growth rate. Additional information from radar or hyperspectral sensors would add extra dimensions to improve the separability of these classes.

For harvest prediction, there was a 12% commission error for the harvested class. To improve this, different spectral indices could be tested and combined to better detect the presence of a sugar cane crop versus no crop. Crop height data from radar or lidar could also be used to determine a drop in crop height, indicating a harvest. Furthermore, some weighting could be applied to account for the fact that 2 year old crops are more likely to be harvested than not.

On the north-east of the study site, a farm on the north-west corner of Lollback Lane and Middle Road on Palmers Island appears to have a disproportionate number of paddocks classified as fallow throughout the decade studied (see Figure 7 and Figures A3-A12). This is atypical for the ratoon cycle of sugarcane crops. Inspecting the area using high resolution data available on Google Earth (Figure 9), it is clear the farm is now composed of

macadamia trees rather than sugarcane crops. Exploring the historical imagery, from 2016 - 2024, the farm was fallow, with grass cover appearing in some years. This results in the classification seen, in particular in 2022 where no paddocks are classified as fallow (see Figure A9).

These issues result from limitations with the training data, as there were no macadamia or grasslands included in the training data. A fourth class could be created, 'other crop', which would allow these paddocks to be excluded from the analysis. However, distinguishing grasslands and macadamia crops from sugar cane crops could prove difficult to do via multispectral data alone. Hyperspectral data could be used to better delineate crop types via spectral signatures, or radar data could be used to identify the straight rows of macadamia trees, as well as crop height.



Figure 9: Farm at corner of Lollback Land and Middle Road, Palmers Island, as viewed via Google Earth high resolution imagery in 2017 and 2025. In 2017, the farm is being prepared for macadamia planting, however it appears no trees were planted until 2025.

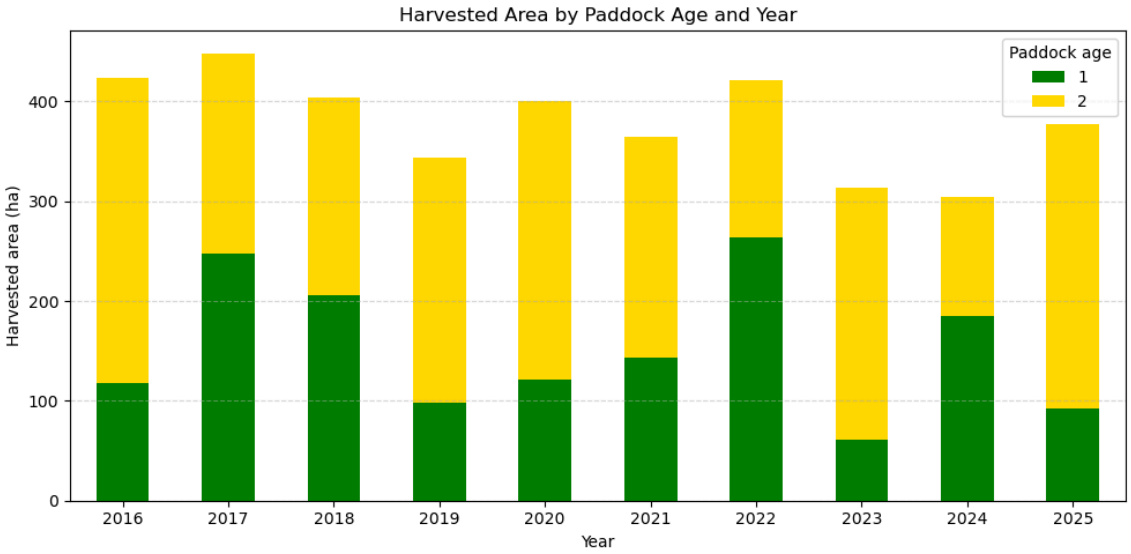


Figure 10: Harvested area per year.

To compare the amount of 1 year old crops harvested against the 2 year old crops harvested, the area was computed using the polygon geometry (Figure 10). As paddock size varies greatly, area is a more useful statistic compared to number. This chart also shows the total amount of area harvested each year for the region.

Figure 10 also shows an approximately 25% decrease in the total area harvested in 2023 and 2024, likely due to the crops lost during the flood experienced by the area in 2022. This flood also impacted farmer's decision making regarding which crops to harvest, resulting in the alternating increase/decrease in 2022-2025 regarding the percentage of 1 year old crop area harvested (Figure 10).

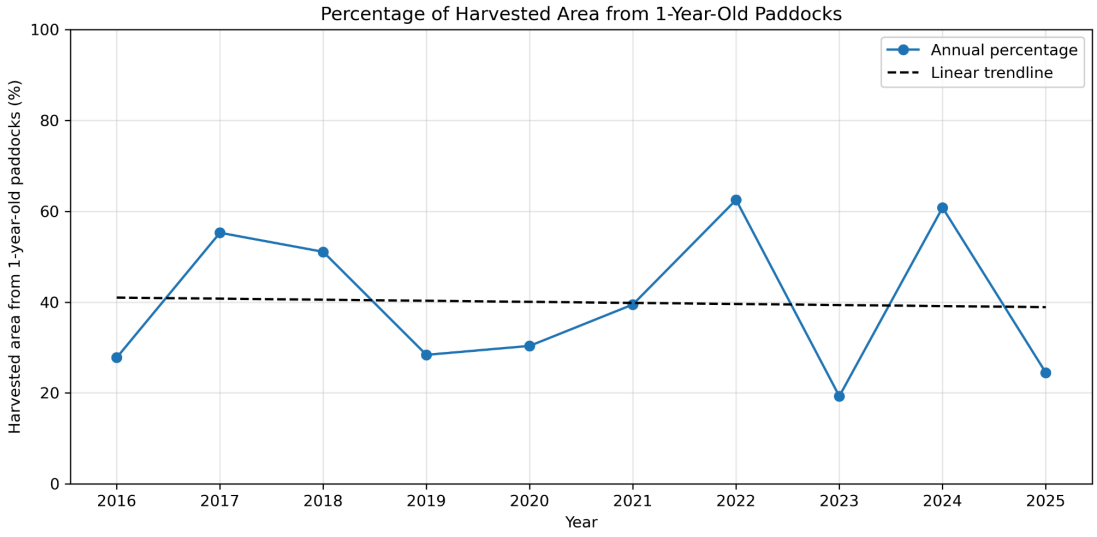


Figure 11: Percentage of harvested area from 1 year old.

As shown in Figure 11, the percentage of area harvested from 1 year old paddocks has remained steady over the last decade, despite a wide array of economic and environmental factors which could influence the farmer's decision making.

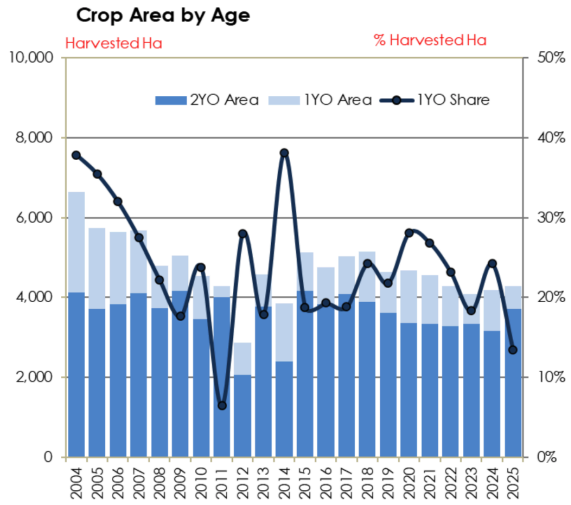


Figure 12: Crop Area by Age for the Harwood Mill Area, reproduced from the Sunshine Sugar 2025 Crop & Productivity Report - Harwood Mill Area (Smith & Ensbey, 2026).

Figure 11 shows an equivalent chart to Figure 9, produced by Smith & Ensbey (2026) using data from the Harwood Sugar Mill for the entire Harwood growing region. The results produced using the remote sensing workflow for the Palmers Channel / Palmers Island region largely agree with the trends observed in the whole region. In particular, the alternating decrease / increase observed in 2023 - 2025 align. For 2016 - 2020, the broader Harwood growing region shows more of an increase in percentage of 1 year old crops than observed in the Palmers Channel / Palmers Island region.

6. Conclusions and Future Work

The decision tree classifier used by the project was able to classify sugarcane crops by age in the Palmers Channel and Palmers Island with an accuracy of 86.36%, using spectral information from Sentinel-2 MSI. To improve the classification accuracy, the following is suggested:

- Use of hyperspectral data to better delineate one year old crops from two year old crops.
- Additional training data for macadamia and grassland paddocks such that they can be classed into 'non-sugar crop' and omitted from further analyses.

- Time-series analysis to understand the phenological cycle, such that age can be determined from time since planted or last harvested.
- Use of radar data to investigate the density of crops, as two year old crops will be denser than one year old.

A simple classification method was used to detect harvest, by calculating the maximum drop in mean NDVI for each paddock over the harvest season. This was found to be 91.74% accurate. This could be improved with the following suggestions:

- Use of other indices and more complex logic to perform the classification.
- Add weighting such that two year old paddocks are almost certain to be harvested.
- Use of radar or lidar data to estimate crop height and factor that into the harvest classification logic.

This project found that classifying sugarcane crops by age is possible using spectral information alone, however additional information is needed to improve the accuracy of the classification.

Future work in this area could include expansion of the workflow to the broader Harwood growing region, and to the other NSW sugar growing regions, Condong and Broadwards. The use of hyperspectral data to classify the different sugarcane varieties planted over the region may provide further insights into trends over the last decade. The use of L-band SAR data could be used to map inundation of sugarcane crops during wet periods, to forecast impact on the harvest in future seasons.

7. Appendix

7.a. Training Data Inspection

The below violin plot shows the distribution of values for each of the three classes. The second plot shows the overlap in the 1 year old and 2 year old classes for the red, red edge 1, green and NIR 1 bands.

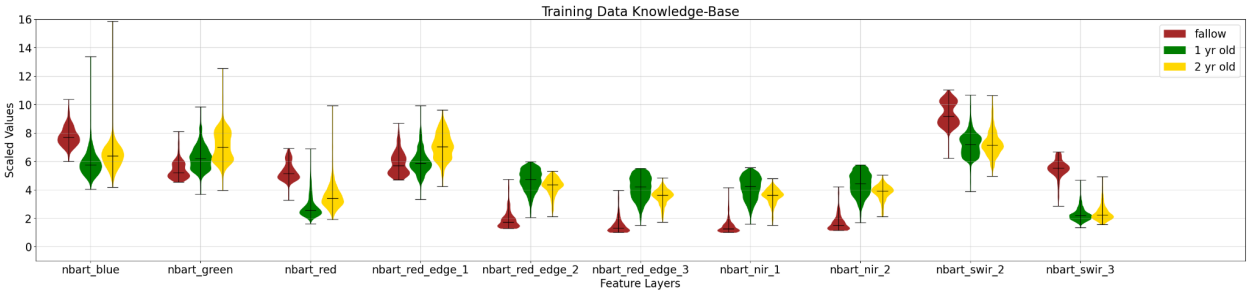


Figure A1: Violin plot of training data for age classification workflow.

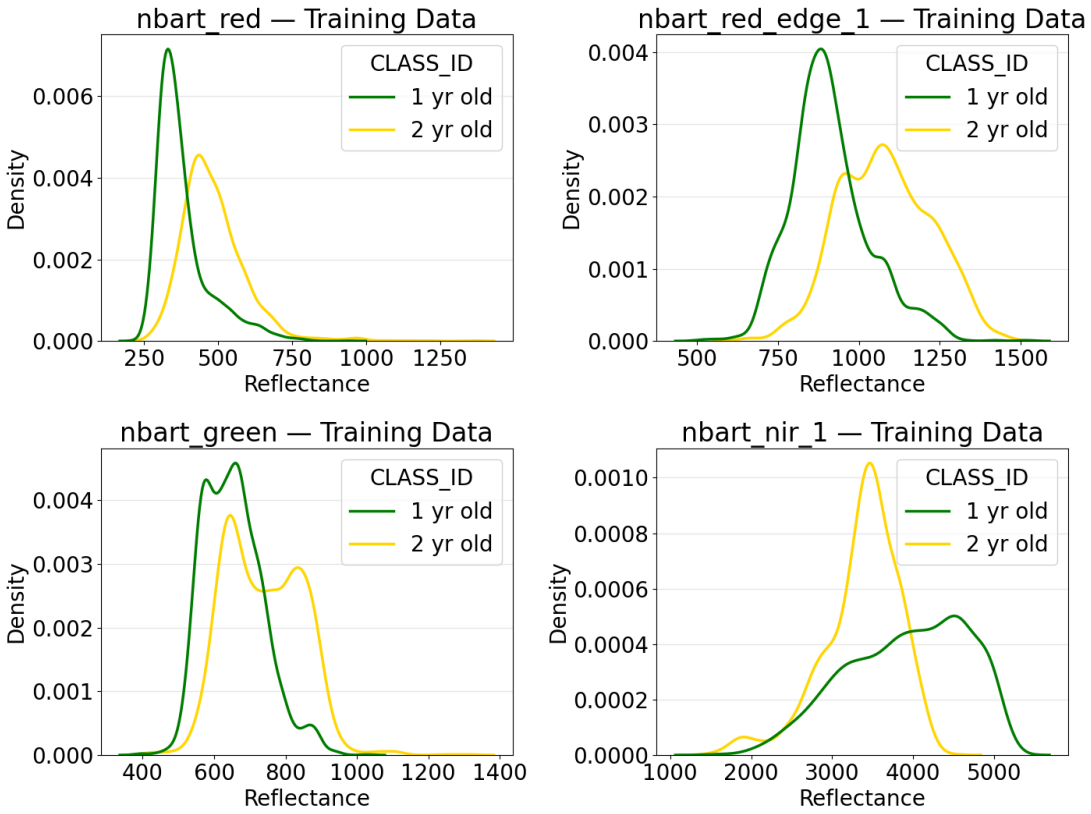


Figure A2: Separation of training data for 1 yr old and 2 yr old classes for red, red edge 1, green and NIR 1 bands.

7.b. Classification Maps, 2016-2025

The below maps show the classification for all years from 2016 - 2025.



Figure A3: Age and harvest classification for sugarcane crops in the Palmers Channel and Palmers Island region, 2016 season.

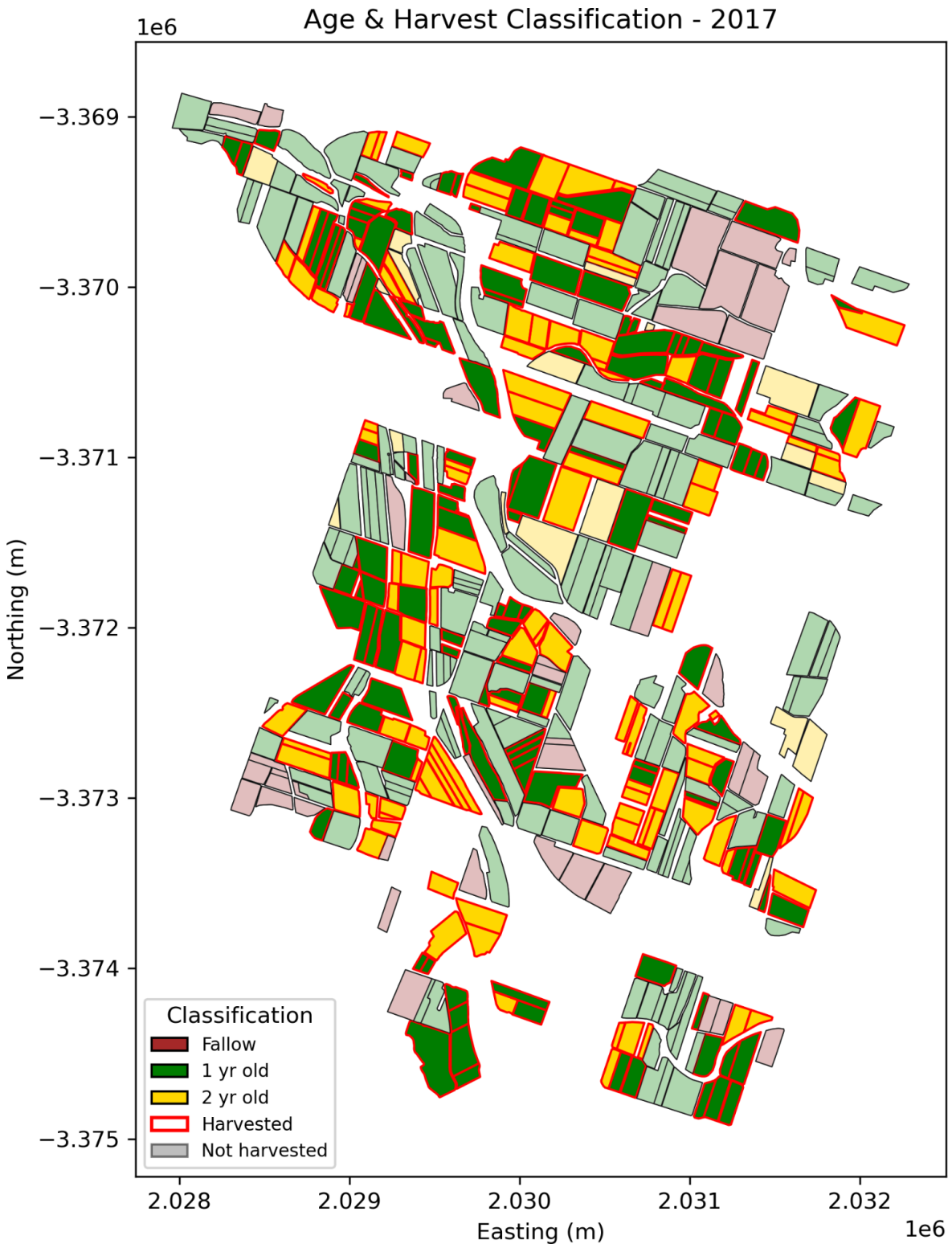


Figure A4: Age and harvest classification for sugarcane crops in the Palmers Channel and Palmers Island region, 2016 season.



Figure A5: Age and harvest classification for sugarcane crops in the Palmers Channel and Palmers Island region, 2018 season.

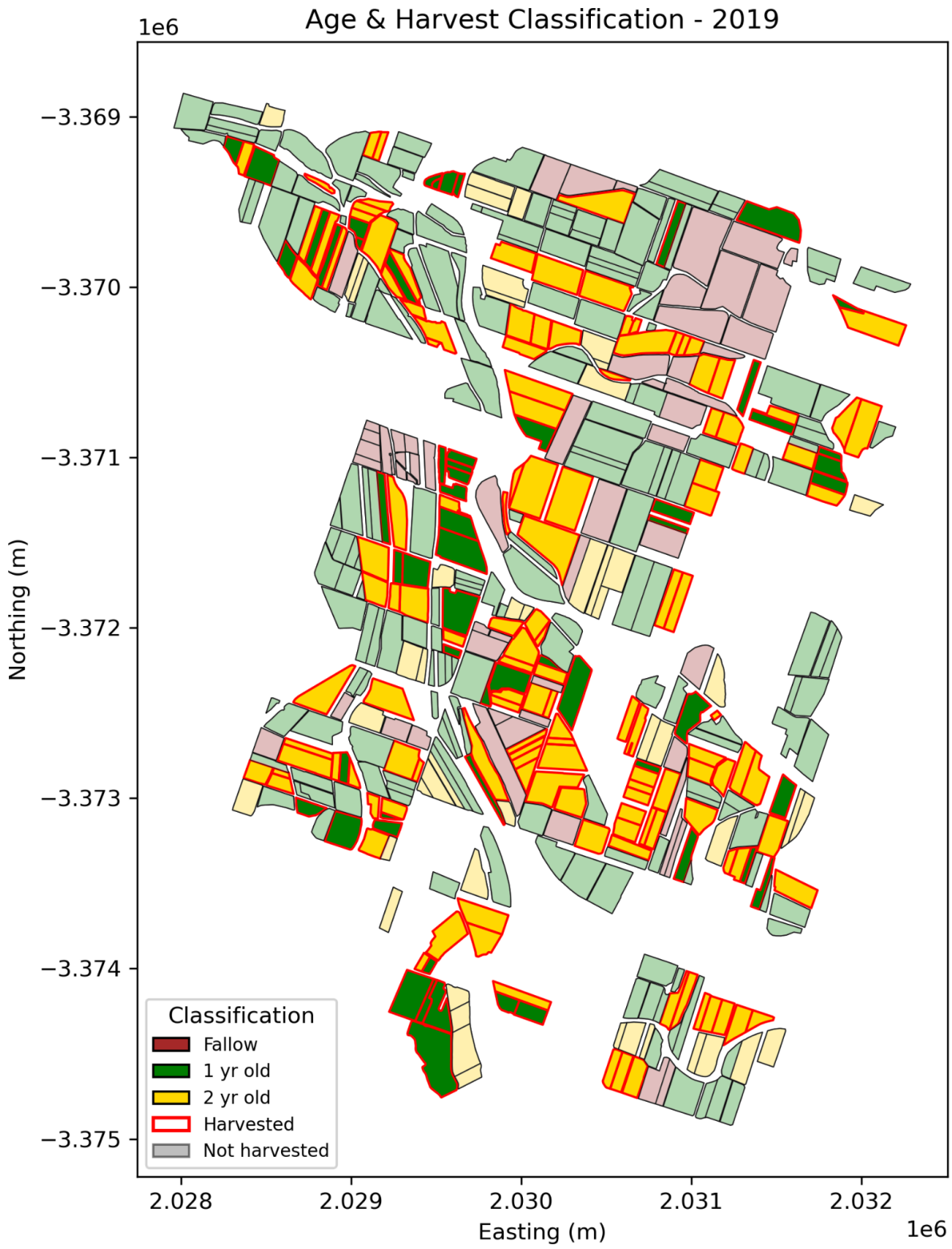


Figure A6: Age and harvest classification for sugarcane crops in the Palmers Channel and Palmers Island region, 2019 season.

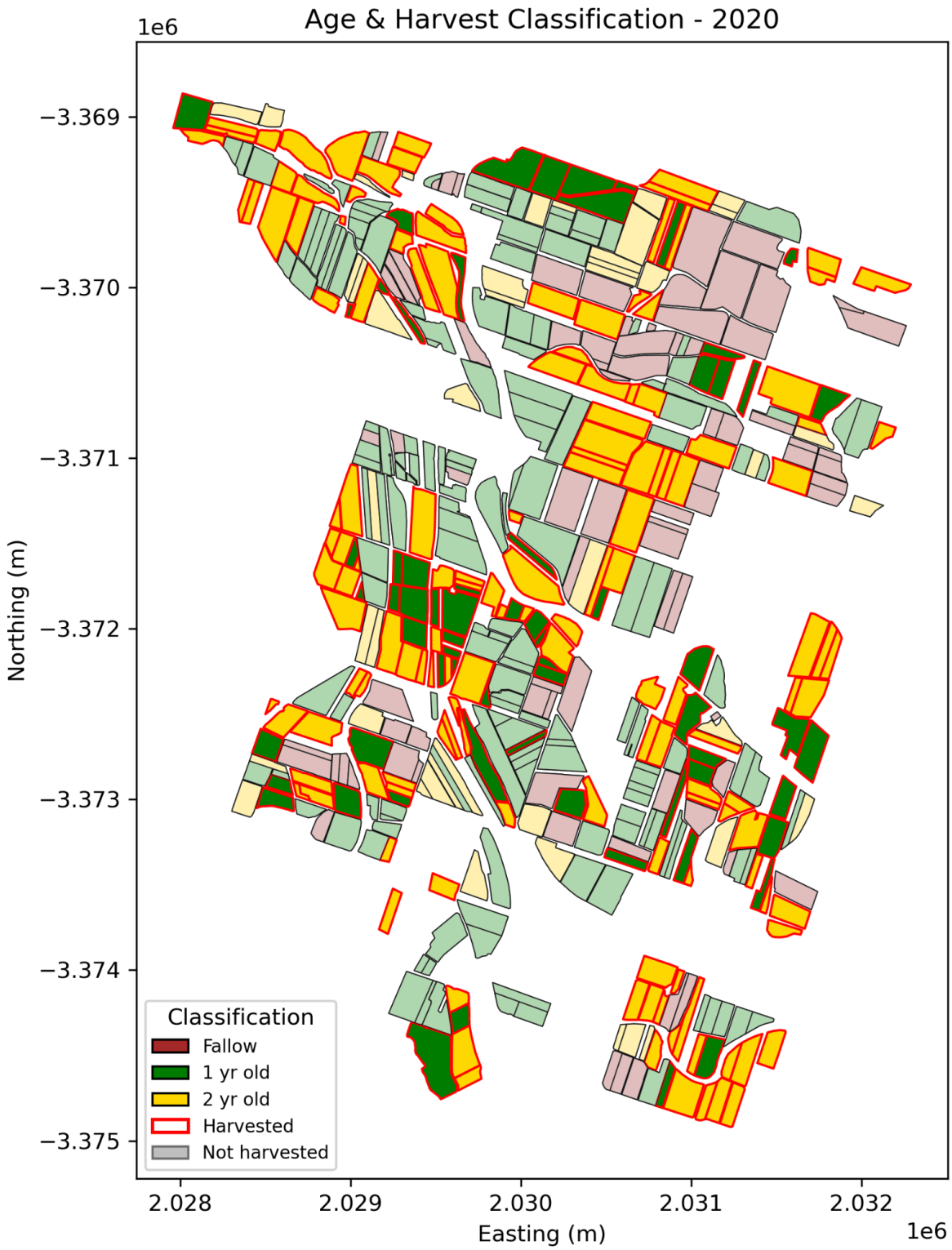


Figure A7: Age and harvest classification for sugarcane crops in the Palmers Channel and Palmers Island region, 2029 season.

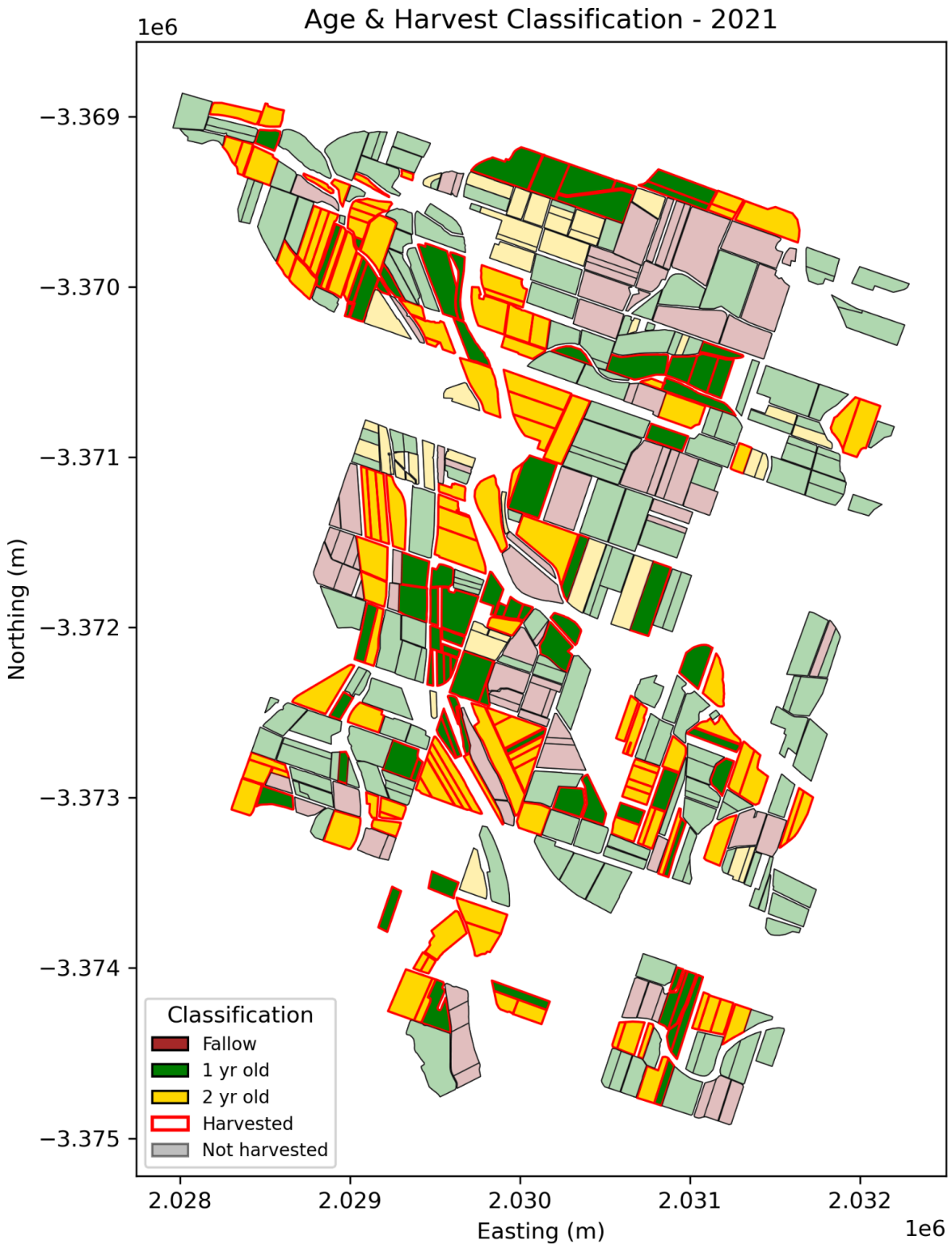


Figure A8: Age and harvest classification for sugarcane crops in the Palmers Channel and Palmers Island region, 2021 season.

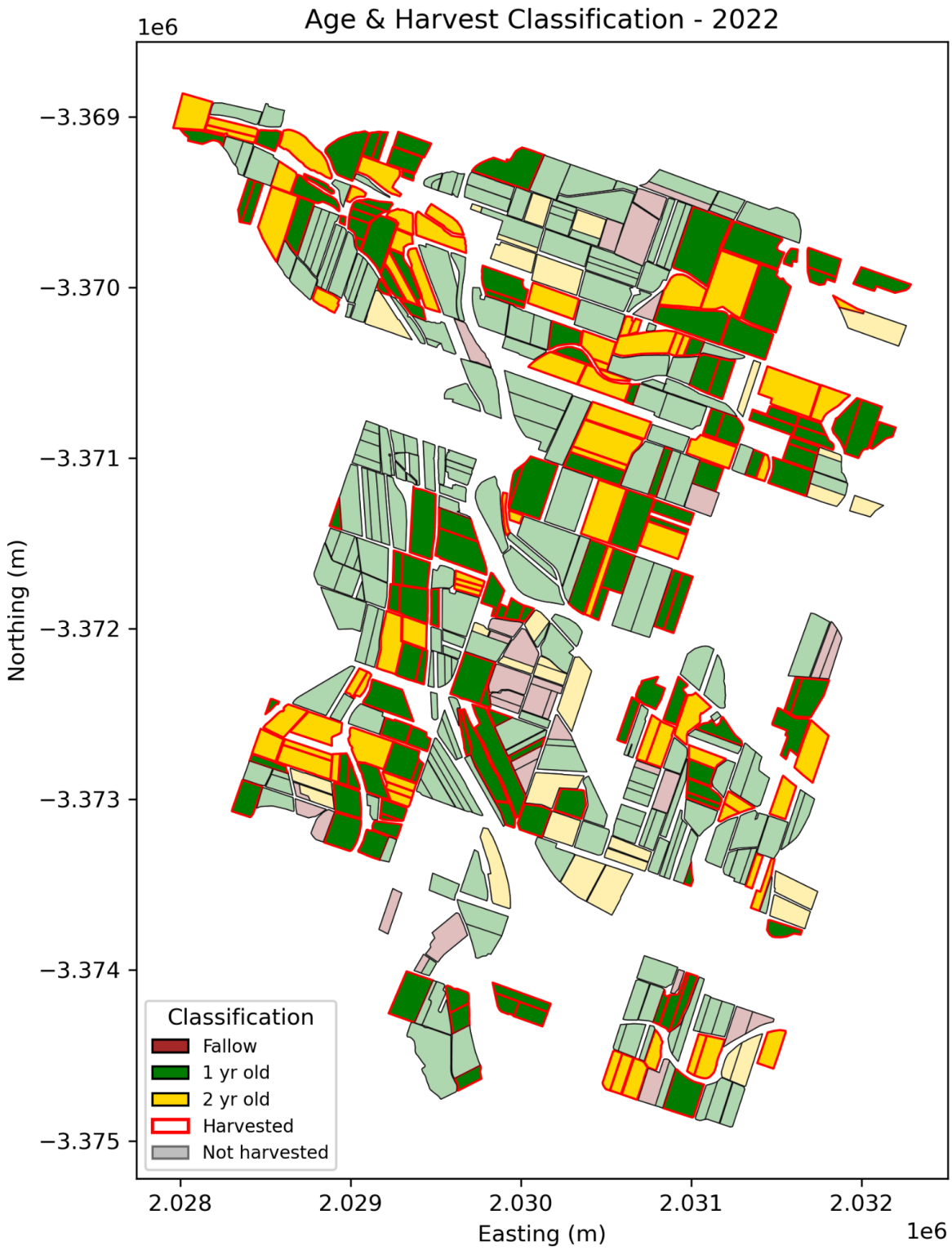


Figure A9: Age and harvest classification for sugarcane crops in the Palmers Channel and Palmers Island region, 2022 season.



Figure A10: Age and harvest classification for sugarcane crops in the Palmers Channel and Palmers Island region, 2023 season.

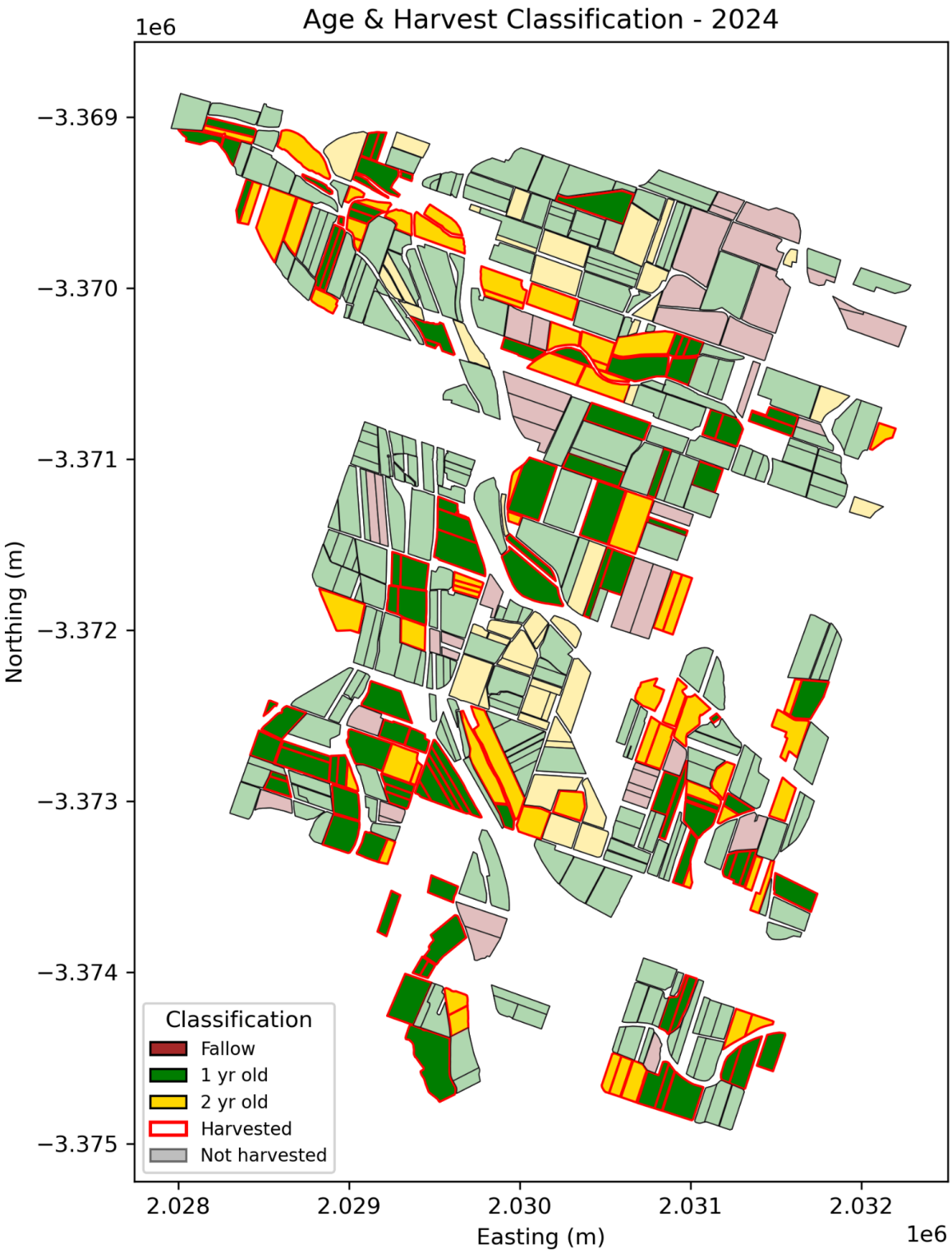


Figure A11: Age and harvest classification for sugarcane crops in the Palmers Channel and Palmers Island region, 2024 season.



Figure A12: Age and harvest classification for sugarcane crops in the Palmers Channel and Palmers Island region, 2025 season.

8. Data Processing Notebooks

All code and ancillary data used for this project can be found in the following GitHub repository: https://github.com/libbyrose/geom7001-sugarcane-age

This includes the following notebooks:

- **01_Plotting_Study_Site.ipynb**: used to load and view the paddock polygons, and filter out the polygons for the study. Produces figures of the study site used in Section 4.a.
- **02_Age_Classification.ipynb**: implements a supervised classification workflow for the age classification, including training the model. Outputs CSV file with classification of polygons by age and year.
- **03_Inspect_training_data.ipynb**: plots Figures A1 and A2 to understand the overlap in training data between classes.
- **04_Harvest_Detection.ipynb**: computes mean NDVI values for each polygon and the drop in NDVI over each season, and classifies as harvested / not harvested. Outputs CSV file with classification of polygons by harvested/not harvested and year.
- **05_Accuracy_Analysis.ipynb**: loads classification results and validation data to compute the overall accuracy as well as the confusion matrices for both harvest and age classification.
- **06_Plotting_and_Statistics.ipynb**: loads classification results to generate the output figures, and compute overall statistics.

9. Generative AI Statement

Generative AI, specifically ChatGPT (OpenAI), was used as an assistive tool throughout the development of this project, largely to support the coding and programming aspects of the work. Support was provided for the following tasks:

- Literature review and research development (as required by the project proposal).
- Brainstorming and refining the research objectives, project title and subtitle.
- Troubleshooting errors in Python to develop workflows to load and process Sentinel-2 surface reflectance data.
- Assistance in generating plots with RGB data and polygons overlaid.
- Understanding GeoPandas and manipulating shapefiles in Python.
- Support in developing an appropriate object-based classification workflow using existing paddock polygons as training units.
- Development of plotting code to compare classes across multiple bands (Figure A2).

- Understanding syntax to calculate and retrieve NDVI values.
- Development of Python code to compare mapped crop ages with reference age information using persistent paddock identifiers.
- Assistance with implementing rules to identify or correct implausible year-to-year crop-age transitions, while accounting for harvest history and missing observations.
- Support for preparing confusion matrices and associated visualisations to assess classification performance (Figure 8).
- Syntax to calculate annual harvested paddock counts and total harvested area in hectares.
- Development of plotting code to compare annual harvest activity by paddock count and area, including separate analyses for one-year-old sugarcane (Figures 9 and 10).
- Assistance with locating DOI links, checking bibliographic details and formatting references in APA 7 style.

Select sections of the prompts and chat log which significantly impacted the project are included in Section 11 below.

10. References

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